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The utility of coronary artery calcium scoring to enhance cardiovascular risk assessment for South Asian adults

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ABSTRACT

South Asian individuals represent a highly diverse population and are one of the fastest growing ethnic groups in the United States. This population has a high prevalence of traditional and non-traditional cardiovascular disease (CVD) risk factors and a disproportionately high prevalence of coronary heart disease. To reflect this, current national society guidelines have designated South Asian ancestry as a "risk enhancing factor" which may be used to guide initiation or intensification of statin therapy. However, current methods of assessing cardiovascular risk in South Asian adults may not adequately capture the true risk in this diverse population. Coronary artery calcium (CAC) scoring provides a reliable, reproducible, and highly personalized method to provide CVD risk assessment and inform subsequent pharmacotherapy recommendations, if indicated. This review describes the utility of CAC scoring for South Asian individuals.

Introduction

South Asian individuals (those with ancestry from Bangladesh, Bhutan, India, the Maldives, Nepal, Pakistan, and Sri Lanka) are among the fastest-growing ethnic groups in the United States (US).[1] This group has been noted to have an increased risk of atherosclerotic cardiovascular disease (CVD;ASCVD) compared to other ethnic groups.[2] A majority of this risk is explainable by traditional risk factors, however, a significant residual risk exists even after accounting for known risk factors. The challenge in accurately estimating risk in this population has led to interest in advanced risk stratification tools that will allow targeted prevention strategies. Coronary artery calcium (CAC) scoring has been shown to be a powerful tool to assess subclinical atherosclerosis and may be particularly helpful in improving the accuracy of risk prediction in the South Asian population. This review describes the role of CAC in enhancing CVD risk assessment for South Asian adults.

CVD in South Asian adults

Various cohort and retrospective studies have demonstrated that the risk of ASCVD in South Asian individuals is approximately ~2× higher compared to non-Hispanic whites (NHW).[3–6] Further, South Asian individuals typically develop ASCVD earlier in life (mean age of 53 years).[7,8] South Asian individuals undergoing cardiac catheterization have been shown to have coronary artery disease (CAD) present at younger ages and have a greater odds of obstructive CAD compared to other racial/ethnic groups.[9] More recent studies examining disaggregated subgroups have shown that the risk of ASCVD is heterogenous among South Asian individuals – the highest risk has been observed among Bangladeshi individuals (HR 3.66, [95% CI: 2.38–5.62]),

Abbreviations: ACC, American College of Cardiology; AHA, American Heart Association; ASCVD, atherosclerotic cardiovascular disease; BMI, Body Mass Index; CAC, coronary artery calcium; CAD, coronary artery disease; CCTA, coronary computed tomography angiography; CHD, coronary heart disease; CI, confidence interval; CVD, cardiovascular disease; GLP-1RA, glucagon-like peptide-1 receptor agonist; hsCRP, high-sensitivity C-reactive protein; HDL, high density lipoprotein; HR, hazard ratio; IBA, inflammatory biomarkers and adipocytokines; LAD, left anterior descending artery; LDL, low density lipoprotein; MACE, major adverse cardiovascular events; MASALA, Mediators of Atherosclerosis in South Asians Living in America; MESA, Multi-Ethnic Study of Atherosclerosis; MI, myocardial infarction; NHW, non-Hispanic whites; NNT, number needed to treat; OR, odds ratio; PCE, pooled cohort equations; RCA, right coronary artery; SBP, systolic blood pressure; SA, South Asian; T2DM, type 2 diabetes mellitus; UK, United Kingdom; US, United States.

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Table 1
Traditional and non-traditional cardiovascular risk factors prevalent in South Asian adults.

Traditional factors	Comments
Obesity/Overweight	SA tend to have higher body fat content for the same BMI when compared with Caucasians. This phenotype has been described as "metabolically-obese-normal weight" SA adults[11,12] Ethnic-specific BMI cut-offs have been proposed for SA adults, with a BMI $\geq$ 23 kg/m^2 considered overweight in this population
Hypertension	Estimated prevalence of 21–43% in SA adults, significantly higher than NHW[13,14] High rates of premature hypertension (highest among Asian Indians [37%] compared to other Asian subpopulations)[7]
Prediabetes and Type 2 diabetes (T2DM)	Rates are heterogenous among SA subgroups, with higher rates seen among those of Bangladeshi origin (10–27%) compared to those originating from Pakistan (11–23%) and India (7–18%)[15–18] SA may have higher rates of insulin resistance and lower beta-cell function.[15] Overall ~2× higher risk of T2DM among SA, with diagnosis occurring ~5 years earlier than NHW.[19,20]
Dyslipidemia	Dyslipidemia among SA individuals is characterized by lower levels of HDL, higher triglycerides, higher total cholesterol, and similar to lower LDL[21,22] Postulated that SA adults carry higher LDL particle burden at a given LDL level (smaller, dense, and more atherogenic LDL)[23,24]
Tobacco	Higher rates of culture-specific forms of tobacco consumption including smokeless tobacco (e.g., paan masala) and cultural smoked products (e.g., hookah) compared to traditional tobacco products[25,26]
Non-traditional factors	**Comments**
Acculturation	SA individuals with stronger traditional cultural beliefs and practices are more likely to have a diet composed of fried snacks, sweets, and high-fat dairy whereas those with weaker beliefs consume more animal protein[27,28] Longer duration of residence in the US is associated with higher CAC[29]
Maternal risk factors	Asian Indian women living in America experience the highest rates of gestational diabetes mellitus and hypertensive disorders of pregnancy[30,31]
Inflammatory markers	C-reactive protein and homocysteine have been shown to be elevated in Asian Indians, though inflammatory markers and adipokines were not found to be associated with subclinical atherosclerosis (assessed by CAC score) in SA adults in the MASALA study[32–34]
Air pollution	High rates of exposure to particulate matter and air pollution, particularly in native SA populations, have been linked to hypertension and cardiovascular death[35,36]

SA – South Asian, BMI – body mass index, NHW – non-Hispanic whites, T2DM – type 2 diabetes mellitus, HDL – high density lipoprotein, LDL – low density lipoprotein, US – United States, CAC – coronary artery calcium, MASALA – Mediators of Atherosclerosis in South Asians Living in America.

followed by Pakistani (HR 2.45, [95% CI: 2.06–2.91]), and Asian Indian (HR 1.83, [95% CI: 1.85–2.33]) individuals.[10] Table 1 highlights traditional and non-traditional risk factors that influence ASCVD risk in South Asian individuals.

CVD risk assessment in South Asian individuals

Given the increased, but heterogenous risk of ASCVD in South Asian individuals, available prevention guidelines have sought strategies to improve the identification of higher risk individuals in this population.

Table 2
Available cardiovascular risk assessment tools and associated guidelines.

Risk Model	Guidelines	Comments
Pooled cohort equations	SA ethnicity is a risk-enhancing factor, and SA individuals at borderline or intermediate risk are recommended statin therapy[37]	May lead to over-estimation of risk in borderline and intermediate risk SA adults when compared to using a CAC score based approach[39]
Modified FRS	2008 NICE Guideline recommends a multiplier of FRS x 1.4 in SA men, no recommendation for SA women[40]	No SA in derivation cohort
SCORE	ESC guidelines recommend multiplying the risk assessed by SCORE by 1.3 for Indians and Bangladeshis, 1.7 for Pakistanis, and 1.1 for other Asian groups[41,42]	No SA in derivation cohort Risk multipliers derived from observed vs. expected risk in the UK[43]
JBS3	2014 Joint British Societies Recommendations on Prevention of Cardiovascular Disease	2.3% SA (1% Indian, 0.5% Pakistani, 0.3% Bangladeshi, and 0.5% Other Asian) in derivation cohort, retrospective studies have shown reasonable performance in identifying high risk SA adults[44,45]
WHO Risk Prediction Charts	None	Has been shown to underestimate risk[46]
QRISK3	None	Despite 2× higher observed ASCVD events among SA adults, risk assessed using QRISK3 was nearly identical for individuals of SA and European ancestry[10]
UKPDS	None	Included 10% Asian-Indian individuals in cohort, used to estimate risk in patients with T2DM[47]

SA – South Asian, CAC – coronary artery calcium, FRS – Framingham Risk Score, NICE – National Institute for Health Care Excellence, SCORE – Systematic Coronary Risk Evaluation, ESC – European Society of Cardiology, UK – United Kingdom, JBS3 – Joint British Society 3, WHO – World Health Organization, ASCVD – atherosclerotic cardiovascular disease UKPDS – UK Prospective Diabetes Study, T2DM – type 2 diabetes.

For example, the 2018 American College of Cardiology (ACC)/American Heart Association (AHA) Guideline on the management of Blood Cholesterol and the 2019 ACC/AHA Guideline on the Prevention of Cardiovascular Disease indicate that "high-risk race/ethnicity" (including South Asian ancestry) is a "risk-enhancing factor" that can be used to guide decisions among adults at borderline (5% to <7.5%) and intermediate (≥7.5% to <20%) risk by the Pooled Cohort Equations (PCE) to revise their 10-year-ASCVD risk estimate and guide statin intensification or initiation.[37,38] This approach may lead to impersonal primary prevention recommendations in the South Asian population, which is characterized by heterogenous ASCVD risk. Designation of race/ethnicity as a high-risk factor may also discourage health-seeking behavior and engagement in care.[2] Further, the AHA/ACC PCE have been shown to inaccurately capture the 10-year risk of ASCVD such that the risk estimated using the PCE was 4.8% for South Asian adults despite an ASCVD rate of 6.8% over a 11-year follow-up period.[10] Though there was a >2-fold increased risk among South Asian individuals in the UK Biobank, the predicted risk using the PCE and the QRISK3 equations was noted to be nearly identical between those of South Asian ancestry and European ancestry.[10] CVD risk assessment tools for use in South Asian individuals are limited considering that available risk algorithms have been derived from limited representation of prospective cohorts of South Asian individuals, do not disaggregate South Asian subgroups, and use crude risk multipliers and adjustments (Table 2).[2]

CAC Scoring to refine ASCVD risk

CAC scoring has been shown to be a powerful predictor of coronary heart disease (CHD) and ASCVD events.[48] The 2018 ACC/AHA Guideline on the Management of Blood Cholesterol and the 2019 ACC/AHA Primary Prevention Guidelines recommend the use of CAC as a shared decision-making aid to guide statin use among adults aged 40–75 who are at borderline or intermediate 10-year-ASCVD risk by the PCE (Class IIa recommendation).[37,38] In patients with a CAC score of zero, treatment with statin therapy can be withheld or delayed with some caveats (tobacco users, those with T2DM, and those with strong family history of premature ASCVD).[37,38] A CAC score of zero is a powerful negative risk predictor, indicating low (<5%) short term (5–10 year) ASCVD risk.[49] If the CAC score is 1–99, the guidelines state it is reasonable to initiate statin therapy in those >55 years old, and if the CAC score is 100 or greater, statin initiation is recommended.[37,38]

CAC scoring may therefore serve as a reliable and reproducible tool to individually assess ASCVD risk and may be of particular benefit in South Asian individuals in whom traditional risk assessment tools have special limitations.

In the Mediators of Atherosclerosis in South Asians Living in America (MASALA) study, South Asian men and women with high (≥7.5%) 10-year risk had significantly higher CAC burden than those with low (<7.5%) 10-year risk. Furthermore, South Asian individuals with high predicted lifetime risk had a significantly increased odds of CAC > 0 (OR for men 1.97, [95% CI: 1.2–3.2] and OR for women 3.14 [95% CI: 1.5–6.6]).[50] However, when studying the performance of the PCE using clinically relevant CAC score categories (0, 1–100, and ≥ 100), South Asian ancestry was associated with higher odds of having CAC = 0 compared to NHW in both the low (<5%) and intermediate (5 to <7.5%) ASCVD risk groups, while South Asian individuals in the high risk group had a similar distribution of CAC compared to NHW.[51] The use of the PCE to routinely assess ASCVD risk in South Asian individuals may therefore lead to overtreatment and over-estimation of risk in low and intermediate ASCVD risk groups. An analysis of MASALA showed that consideration of borderline (5 to <7.5%) and intermediate (7.5 to <20%) ASCVD risk individuals as statin candidates would lead to 35% of participants with CAC = 0 being recommended for statin therapy.[39] Data from the South Asian Health Center's AIM to Prevent Program showed that if ASCVD risk had been only assessed using the PCE, 21% of participants who had CAC > 0 but ASCVD risk scores <7.5% would not have been offered statin therapy that may have been indicated.[52] ASCVD risk scoring also misclassified 49% of those CAC = 0 but with ASCVD risk ≥7.5% in whom statin therapy would have no benefit.[52] These findings highlight that CAC scoring may have powerful clinical utility in South Asian individuals at borderline and intermediate risk in whom the PCE have been shown to be potentially inaccurate and lead to both under-treatment and over-treatment. Alternatively, the PCE have shown reasonable utility in those at low (<5%) and high (>20%) ASCVD risk. It is important to note that longitudinal data of ASCVD events in South Asian adults in the US is sparse, and further data are needed to validate the prognosis of CAC = 0 in this population and allow for re-calibration of current risk assessment tools.

An analysis of MASALA and MESA showed that age-adjusted CAC incidence is similar in South Asian men compared with white, Black, and Latino men, but higher than in Chinese men (11.1% vs 5.7% incidence). After adjustment for traditional risk factors, Chinese, Black, and Latino men had significantly less CAC change compared to South Asian men, though there was no difference between South Asian and white men and no difference in CAC incidence or progression between South Asian women and women of other racial/ethnic groups.[53] Predictors of change in CAC in South Asian adults included older age, male sex, tobacco use, statin use, hypertension medication use, and T2DM.[53] Direct measurement of CAC in the MASALA cohort with serial CT showed that incident CAC was most common in the left anterior descending (LAD) artery and the greatest progression in CAC volume was in the right coronary artery (RCA). This is additive to coronary computed tomography angiography (CTA) and cardiac catheterization data, which has shown that South Asian adults have more aggressive and diffuse arterial calcifications and more severe CAD in the proximal LAD and RCA segments compared to Caucasians.[54,55] South Asian adults have been found to have lower CAC volume but similar CAC density compared to NHW, and similar CAC volumes compared to non-whites.[56] ASCVD risk estimated by the PCE has been associated with CAC volume and density in South Asian individuals, and HDL is directly associated with CAC density while waist circumference is inversely associated with it in subjects in MASALA.[57] Smoking has been associated with CAC volume progression, while lipoprotein (a) and exercise were associated with progression of peak CAC density in South Asian adults.[58]

It is important to note that traditional risk factors only partially explain the risk of ASCVD in South Asian individuals suggesting possible familial and genetic influences. An analysis of MASALA showed that having a family history of CHD in South Asian individuals is associated with severe subclinical atherosclerosis (CAC > 300).[59] Inflammatory biomarkers and adipocytokines (IBA) have been shown to contribute to atherosclerosis by promoting inflammation of the vasculature. Though women in the MASALA cohort had significantly higher levels of high-sensitivity C-reactive protein (hsCRP), adiponectin, and leptin and significantly lower levels of TNF-α than men, there was no association between these biomarkers and levels of CAC.[32] Further, addition of IBA to the PCE did not improve discrimination of CAC presence or severity.[32]

Clinical applications of CAC scoring in South Asian individuals

Measurement of CAC score in South Asian individuals may allow for enhanced risk assessment, improve allocation of primary prevention therapies, and help guide lifestyle management recommendations.

An analysis of hypertension guidelines across CAC quintiles showed that a significantly higher proportion of South Asian individuals were diagnosed with hypertension and recommended anti-hypertensive therapy according to the 2017 ACC/AHA guidelines compared to the Joint National Committee on Prevention, Detection, Evaluation, and Treatment of High Blood Pressure (JNC7) guidelines.[60] This discordance was higher among participants with CAC > 100, and CAC may therefore allow for assistance in allocating antihypertensive therapy. CAC score may also be used to set specific blood pressure targets – an analysis of the MESA showed that CAC may allow for personalized blood pressure goals in patients with systolic blood pressure (SBP) 120–159 mmHg and ASCVD risk <15%.[61] For example, CAC may aid in the decision to choose between a traditional SBP goal of 140 mmHg or a more intensive goal of 120 mmHg in individuals with ASCVD risk estimated at 5–15%.[61]

Due to the direct relationship between CAC score and ASCVD, National Lipid Association guidelines have proposed LDL-C targets based on baseline CAC.[62] The 2022 ACC Expert Consensus Decision Pathway on the Role of Non-statin Therapies for LDL-C Lowering in the Management of ASCVD Risk sets LDL-C goals based on baseline CAC and percentile. For example, those with CAC of 1–99 and < 75th percentile for age/sex/race may be initiated on moderate-intensity statin therapy to target a 30–49% reduction in LDL-C and LDL-C < 100 mg/dL.[63] If this goal is not achieved, intensification of statin therapy can be considered. In those with higher baseline CAC score (e.g., >100 Agatston units), guidelines propose greater LDL-C reduction and recommend the addition of non-statin therapies such as ezetimibe and PCSK9 inhibitors.[63] In patients without clinical ASCVD, T2DM, or LDL-C ≥ 190 mg/dL (e.g., adults for whom there is uncertainty or patient hesitancy to initiate statin therapy), the CAC score allows for personalized recommendations for strength of statin therapy and LDL-C targets.

Similarly, CAC may allow risk refinement in South Asian patients with T2DM.[64] This is especially applicable to South Asian adults, who have disproportionately higher rates of prediabetes and T2DM, and T2DM at lower levels of adiposity and at a younger age.[19,20] An analysis of the MESA, the Coronary Artery Risk Development in Young Adults

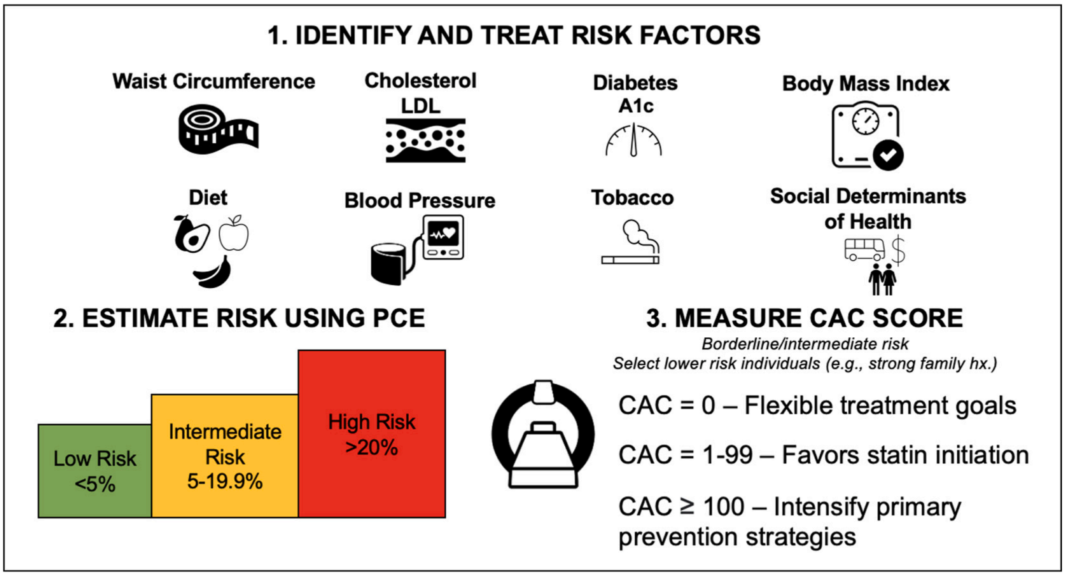

Fig. 1. Summary of use of coronary artery calcium in risk assessment and allocation of statin therapy in South Asian individuals. CAC may be used in cases of intermediate or borderline risk in whom the role for primary prevention therapies is unclear, or in those who are younger with a strong family history of ASCVD or multiple risk factors. CAC = 0 permits flexible treatment goals and any coronary calcium (CAC >0) favors lifestyle optimization and identification and management of other risk factors; CAC 1–99 favors statin initiation; CAC ≥ 100 suggests need to intensify primary prevention pharmacotherapy (e.g., aspirin, statin).
PCE – pooled cohort equations, CAC – coronary artery calcium, LDL – low density lipoprotein.

Table 3
Recommendations for applications of CAC scoring in South Asian individuals.

Clinical scenarios in which to consider CAC scoring in South Asian adults
1. Individuals at borderline and intermediate risk by the PCE.
2. Individuals at low risk by the PCE but with a family history of premature CAD.[67]
3. Younger adults (<45 years) with multiple traditional risk factors such as hypertension, diabetes, obesity, tobacco use, or dyslipidemia.

CAC – coronary artery calcium, PCE – pooled cohort equations, CAD – coronary artery disease.

(CARDIA) study, and the Dallas Heart Study (DHS) examined the role of CAC in allocation of glucagon-like peptide 1 receptor agonists (GLP-1RA) in individuals of other ethnic groups.[65]

The routine use of aspirin for primary prevention has recently been called into question, although CAC may help adjudicate its role. An analysis of the MESA cohort showed that individuals with CAC > 100 may be the best candidates for aspirin therapy, and patients with CAC = 0 would have a higher risk of bleeding than potential benefit from aspirin.[66] CAC can serve as a tool to identify the patients most likely to benefit from aspirin therapy (e.g. CAC >100 without bleeding contraindications), as endorsed by the Society of Cardiovascular Computed Tomography (SCCT).[67]

Fig. 1 illustrates a proposed framework highlighting integration of the CAC score with traditional risk factor modification in South Asian individuals based on available data.[68,69] An approach to selecting South Asian adults who may benefit from CAC scoring is illustrated in Table 3. Though current CAC scoring recommendations begin at age 40, earlier CAC scoring in South Asian adults may have utility due to the earlier development of ASCVD in this group compared to other racial/ethnic groups, particularly among those with multiple traditional risk factors.[67,70] Further, SCCT recommends that individuals with a family history of premature ASCVD but deemed low risk by the PCE may also benefit from CAC scoring.[67]

Coronary CTA in South Asian individuals

While cohort studies have examined CAC among South Asian adults, coronary CTA (CCTA) is an understudied tool. CCTA can help identify individual plaque characteristics and high-risk plaque characteristics such as positive remodeling, spotty calcification, napkin ring sign, and low attenuation plaque, which may allow for enhanced risk assessment beyond the CAC score.[71] Cross-sectional analysis of the Miami Heart Study (MiHeart) showed that 16% of individuals with CAC = 0 had any plaque on CCTA, though only 0.8% of participants with CAC = 0 had stenosis ≥50% and 2.3% had high-risk features. Male sex, overweight, and obesity were independent predictors of plaque among those with CAC = 0.[72] In asymptomatic patients in the Swedish Cardiopulmonary Bioimage Study (SCAPIS), all those with CAC > 400 had atherosclerosis seen on CCTA and among those with CAC = 0, 5.5% had atherosclerosis and 0.4% had significant stenosis.[73] Major society guidelines have not yet endorsed coronary CTA for ASCVD risk assessment in asymptomatic patients, however, a clinical practice statement from the American Society for Preventive Cardiology suggests CCTA may be used in South Asian individuals with a strong family history, family history of premature ASCVD, those with diabetes, and those who smoke tobacco, among others.[74]

Quantitative differences in atherosclerotic plaque burden in South Asian adults have been shown using CCTA – a study of 543 South Asian and 420 Caucasian patients with chest pain undergoing CCTA showed that the percentage of non-calcified plaque composition was lower in Caucasians compared to South Asians.[75] Compared to Caucasian patients, South Asian individuals have been shown to have higher prevalence of significant CAD and more frequent involvement of the LAD using CCTA.[76] Studies examining coronary CTA in South Asian subjects are summarized in Table 4. Further longitudinal research is required to assess the utility of CCTA in South Asian individuals. To this end, the South Asians and Coronary Plaque Registry plans to study the number and type of adverse and protective CCTA-derived plaque features

Table 4
Studies examining the use of coronary CTA in South Asian individuals.

Authors	Cohort	Population	Findings
Koulaouzidis et al.[54]	Institutional data from the UK	101 SA patients with angina referred for CCTA	SA with chest pain had a higher number of diseased coronary segments and more aggressive and diffuse arterial calcification compared to Caucasians.
Hasan et al.[55]	Institutional data the US	63 SA men with acute coronary syndrome, or chronic coronary syndrome	SA undergoing quantitative coronary angiography had more severe CAD, smaller proximal LAD lumen, higher percent stenosis per vessel, and higher percentage of multivessel disease compared to Caucasians.
Villadsen et al.[75]	Institutional data from the UK	543 SA patients (297 male) with stable angina in a non-acute setting	SA had higher non-calcified plaque composition compared to Caucasians (90.42% vs. 80.95%, $p < .001$).
Roos et al.[76]	Institutional data from the Netherlands	120 asymptomatic SA patients with T2DM	SA had higher CAC and higher prevalence of significant CAD (>50% stenosis) involving more coronary vessels and segments compared to matched Caucasians. Significant CAD most frequent in the LAD among SA
Tillin et al.[77]	Institutional data from the UK	41 SA men with established CAD	SA undergoing quantitative coronary angiography had more proximal LAD stenosis and narrower LAD diameter.
Manubolu et al.[78]	Institutional data from the US	104 SA who underwent CCTA	SA had higher calcified percentage atheroma volume and higher prevalence of obstructive disease (defined as >50% stenosis) compared to NHW
Ramjattan et al.[79]	Institutional data from the US	262 SA who underwent CCTA	No difference between CAC score, plaque burden, or diffusivity index between SA and NHW

SA – South Asian, CAD – coronary artery disease, LAD – left anterior descending artery, UK – United Kingdom, T2DM – type 2 diabetes mellitus, CAC – coronary artery calcium, CCTA – coronary computed tomography angiography.

(ClinicalTrials.gov: NCT05367297) in South Asians in the US while the Pakistan Study of Premature Coronary Atherosclerosis in Young Adults (PAKSEHAT) is studying the burden of atherosclerotic plaque assessed via coronary CTA in Pakistan (ClinicalTrials.gov: NCT05156736).

Future directions

Further data is needed to identify the role of CAC scoring in younger (<45 years) South Asian individuals and to verify the prognosis of CAC = 0 in this group. Additionally, the "warranty period" of CAC = 0 should be further studied specifically in South Asian individuals, to determine when it may be reasonable to rescan patients with CAC = 0. A 3–5-year interval for rescanning has been proposed for individuals at borderline or intermediate risk and a 6–7-year interval for those at low risk, however, recommendations for routine repeat CAC testing in South Asian individuals have not been specified in existing prevention guidelines.[37,49] Importantly, in a study of 698 South Asian adults over a mean 4.8 follow-up period, the age-adjusted CAC incidence (among those with no CAC at baseline) was 8.8% [95% CI: 6.8–10.8%] in men and 3.6% [95% CI: 2.5–4.8%] in women.[53] Further, considering that South Asian men and NHW men have similar levels of CAC, the MESA 10-Year CHD risk calculator with CAC score may be of utility to refine risk, at least for South Asian men, though validation of this method is required before implementation into practice.[80]

Conclusion

South Asian individuals have a disproportionately higher risk of ASCVD, driven largely by traditional risk factors such as metabolic syndrome, T2DM, hypertension, and dyslipidemia commonly found in this population. However, risk can be highly heterogenous. Notably, most of the literature has focused on Asian Indians, as such, further prospective data examining disaggregated South Asian subpopulations (including native and migrant populations, second and third generation subgroups, and ethnically mixed families, for example) is needed to identify and manage ASCVD risk, particularly when the prevalence of many traditional risk factors is higher. Current risk algorithms may inaccurately capture ASCVD risk in this population, particularly those classified as intermediate/borderline risk. Hence, the use of CAC may help navigate this shortcoming, allowing for personalized risk assessment and therapeutic considerations.

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Declaration of competing interest

None.

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